Bayesian Modelling Exercises

Exercise 1: Flu Diagnosis

Problem:
A patient goes to the clinic with a **fever**. Doctors know the following:

- 5% of patients have **Flu**: (P(Flu) = 0.05)
- 10% of patients have **Cold**: (P(Cold) = 0.10)
- The probability of **fever given Flu**: (P(Fever | Flu) = 0.9)
- The probability of **fever given no Flu**: (P(Fever | \neg Flu) = 0.1)

Question: What is the probability that the patient has **Flu given that they have a fever**?

Solution: Using Bayes' Rule

Bayes' Rule formula:

$$P(Flu|Fever) = \frac{P(Fever|Flu) \cdot P(Flu)}{P(Fever)}$$

Step 1: Compute total probability of Fever

$$P(Fever) = P(Fever|Flu) \cdot P(Flu) + P(Fever|\neg Flu) \cdot P(\neg Flu)$$

$$P(Fever) = (0.9)(0.05) + (0.1)(0.95) = 0.045 + 0.095 = 0.14$$

Step 2: Apply Bayes' Rule

$$P(Flu|Fever) = \frac{0.9 \cdot 0.05}{0.14} = \frac{0.045}{0.14} \approx 0.321$$

✅ **Answer:** ~32.1%

Exercise 2: Disease Test

Problem:
A rare disease affects 1% of the population. A diagnostic test for the disease has:

- Sensitivity (true positive rate): 95%
- Specificity (true negative rate): 90%

A patient tests **positive**. What is the probability they **actually have the disease**?

Solution

Let D = disease, T = positive test.

Step 1: Bayes' Rule

$$P(D|T) = \frac{P(T|D) \cdot P(D)}{P(T)}$$

Step 2: Compute total probability of positive test

$$P(T) = P(T|D)P(D) + P(T|\neg D)P(\neg D)$$

- P(T | D) = 0.95
- P(T | ¬D) = 1 - specificity = 1 - 0.9 = 0.1
- P(D) = 0.01, P(¬D) = 0.99

$$P(T) = (0.95)(0.01) + (0.1)(0.99) = 0.0095 + 0.099 = 0.1085$$

Step 3: Compute posterior probability

$$P(D|T) = \frac{0.95 \cdot 0.01}{0.1085} = \frac{0.0095}{0.1085} \approx 0.0875$$

✅ **Answer:** ~8.75%

> Even with a positive test, the probability is still low because the disease is rare. This is a classic **base rate fallacy** example.

Exercise 3: Spam Email

Problem:
Suppose 20% of emails are spam. A spam filter detects certain keywords:

- P(Keyword | Spam) = 0.9
- P(Keyword | Not Spam) = 0.05

If an email contains the keyword, what is the probability it is spam?

Solution

Step 1: Bayes' Rule

$$P(Spam|Keyword) = \frac{P(Keyword|Spam) \cdot P(Spam)}{P(Keyword)}$$

Step 2: Compute P(Keyword)

$$P(Keyword) = P(Keyword|Spam)P(Spam) + P(Keyword|\neg Spam)P(\neg Spam)$$

$$P(Keyword) = (0.9)(0.2) + (0.05)(0.8) = 0.18 + 0.04 = 0.22$$

Step 3: Compute posterior probability

$$P(Spam|Keyword) = \frac{0.9 \cdot 0.2}{0.22} = \frac{0.18}{0.22} \approx 0.818$$

✅ **Answer:** ~81.8%

Exercise 1: Flu and Symptoms

Scenario (Word Problem):
A clinic monitors patients for **Flu (F)**, **Fever (Fe)**, and **Cough (Co)**. Doctors provide the following probabilities:

Priors:

- P(F=True) = 0.05
- P(F=False) = 0.95

Conditional probabilities:

- Fever depends on Flu:

F	P(Fe=True)	P(Fe=False)
T	0.9	0.1
F	0.1	0.9

- Cough depends on Flu:

F	P(Co=True)	P(Co=False)
T	0.8	0.2
F	0.2	0.8

Task: Compute the **joint probability distribution** for all combinations of F, Fe, Co.

Solution

Step 1: List all possible combinations of F, Fe, Co

F	Fe	Co
T	T	T
T	T	F
T	F	T
T	F	F
F	T	T
F	T	F
F	F	T
F	F	F

Step 2: Apply BN factorization

$$P(F, Fe, Co) = P(F) \cdot P(Fe|F) \cdot P(Co|F)$$

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Step 3: Compute joint probabilities

1. F=T, Fe=T, Co=T:

$$P = 0.05 \cdot 0.9 \cdot 0.8 = 0.036$$

2. F=T, Fe=T, Co=F:

$$P = 0.05 \cdot 0.9 \cdot 0.2 = 0.009$$

3. F=T, Fe=F, Co=T:

$$P = 0.05 \cdot 0.1 \cdot 0.8 = 0.004$$

4. F=T, Fe=F, Co=F:

$$P = 0.05 \cdot 0.1 \cdot 0.2 = 0.001$$

5. F=F, Fe=T, Co=T:

$$P = 0.95 \cdot 0.1 \cdot 0.2 = 0.019$$

6. F=F, Fe=T, Co=F:

$$P = 0.95 \cdot 0.1 \cdot 0.8 = 0.076$$

7. F=F, Fe=F, Co=T:

$$P = 0.95 \cdot 0.9 \cdot 0.2 = 0.171$$

8. F=F, Fe=F, Co=F:

$$P = 0.95 \cdot 0.9 \cdot 0.8 = 0.684$$

✅ Step 4: Joint probability table

F	Fe	Co	P(F,Fe,Co)
T	T	T	0.036
T	T	F	0.009
T	F	T	0.004
T	F	F	0.001
F	T	T	0.019
F	T	F	0.076
F	F	T	0.171
F	F	F	0.684

All probabilities sum to 1: 0.036+0.009+0.004+0.001+0.019+0.076+0.171+0.684 = 1 ✅

Exercise 2: Weather Example

Scenario:
Variables: **Rain (R)**, **Sprinkler (S)**, **Wet Grass (W)**

Dependencies:

- Rain influences Sprinkler: Sprinkler independent of Rain in this case.
- Wet Grass depends on both Rain and Sprinkler.

Priors:

- P(R=T) = 0.2, P(R=F) = 0.8
- P(S=T) = 0.5, P(S=F) = 0.5

Conditional probabilities for Wet Grass:

R	S	P(W=T)	P(W=F)
T	T	0.99	0.01
T	F	0.8	0.2
F	T	0.9	0.1
F	F	0.0	1.0

Task: Compute P(R,S,W) for all combinations.

Solution

Solution

Factorization:

$$P(R, S, W) = P(R) \cdot P(S) \cdot P(W|R, S)$$

Compute some examples:

1. R=T, S=T, W=T:

$$P = 0.2 \cdot 0.5 \cdot 0.99 = 0.099$$

2. R=T, S=T, W=F:

$$P = 0.2 \cdot 0.5 \cdot 0.01 = 0.001$$

3. R=T, S=F, W=T:

$$P = 0.2 \cdot 0.5 \cdot 0.8 = 0.08$$

4. R=T, S=F, W=F:

$$P = 0.2 \cdot 0.5 \cdot 0.2 = 0.02$$

5. R=F, S=T, W=T:

$$P = 0.8 \cdot 0.5 \cdot 0.9 = 0.36$$

6. R=F, S=T, W=F:

$$P = 0.8 \cdot 0.5 \cdot 0.1 = 0.04$$

7. R=F, S=F, W=T:

$$P = 0.8 \cdot 0.5 \cdot 0.0 = 0.0$$

8. R=F, S=F, W=F:

$$P = 0.8 \cdot 0.5 \cdot 1.0 = 0.4$$

✅ Joint probability table:

R	S	W	P(R,S,W)
T	T	T	0.099
T	T	F	0.001
T	F	T	0.08
T	F	F	0.02
F	T	T	0.36
F	T	F	0.04
F	F	T	0.0
F	F	F	0.4